

Q3 Solve the difference eqn

$$x[n] = \frac{2}{3} x[n-1] - \frac{1}{9} x[n-2] + \left(\frac{1}{2}\right)^n$$

with initial conditions $x[-1] = 1$, $x[-2] = -1$ using z-transform technique.

Sol Taking z-transform of both sides, we obtain,

$$X(z) = \frac{2}{3} (z^{-1} X(z) + \underbrace{x[-1]}_1) - \frac{1}{9} (z^{-2} X(z) + z^{-1} \underbrace{x[-1]}_1 + \underbrace{x[-2]}_{-1}) + \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$\Rightarrow X(z) \left[1 - \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2} \right] = \frac{2}{3} + \frac{1}{9} - \frac{z^{-1}}{9} + \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$\Rightarrow X(z) = \frac{16 - \frac{9}{2} z^{-1} + \frac{1}{2} z^{-2}}{9(1 - \frac{1}{3} z^{-1})^2 (1 - \frac{1}{2} z^{-1})} = \frac{A}{(1 - \frac{1}{3} z^{-1})} + \frac{B}{(1 - \frac{1}{3} z^{-1})^2} + \frac{C}{1 - \frac{1}{2} z^{-1}}$$

$$\Rightarrow \begin{aligned} \frac{1}{6} A + \frac{1}{9} C &= \frac{1}{18} \\ -\frac{5}{6} A - \frac{B}{2} - \frac{2}{3} C &= -\frac{1}{2} \\ A + B + C &= \frac{16}{9} \end{aligned} \Rightarrow \begin{bmatrix} \frac{1}{6} & 0 & \frac{1}{9} & 1 & \frac{1}{18} \\ -\frac{5}{6} & -\frac{1}{2} & -\frac{2}{3} & 1 & -\frac{1}{2} \\ 1 & 1 & 1 & 1 & \frac{16}{9} \end{bmatrix} \xrightarrow{\text{e.r.o.}} \begin{bmatrix} 3 & 0 & 2 & 1 \\ 0 & -9 & -2 & -4 \\ 0 & 0 & 2 & 8 \end{bmatrix}$$

$$\Rightarrow A = -17/3 \quad C = 9 \quad B = -14/9$$

$$\Rightarrow X(z) = -\frac{17}{3} \cdot \frac{1}{1 - \frac{1}{3} z^{-1}} - \frac{14}{9} \frac{1}{(1 - \frac{1}{3} z^{-1})^2} + 9 \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$\Rightarrow x[n] = -\frac{17}{3} \left(\frac{1}{3}\right)^n - \frac{14}{9} (n+1) \left(\frac{1}{3}\right)^n + 9 \left(\frac{1}{2}\right)^n$$

$$\text{OR } x[n] = -\frac{65}{9} \left(\frac{1}{3}\right)^n - \frac{14}{9} n \left(\frac{1}{3}\right)^n + 9 \left(\frac{1}{2}\right)^n$$

Q4 solve $x'' - x' - 2x = e^{-2t}$ with initial conditions $x(0) = 1$, $x'(0) = 1$ using Laplace Transform technique.

Soln Take Laplace Transform of both sides,

$$s^2 X(s) - s \underbrace{x(0)}_1 - \underbrace{x'(0)}_1 - (s X(s) - \underbrace{x(0)}_1) - 2X(s) = \frac{1}{s+2}$$

$$X(s) [s^2 - s - 2] = \frac{1}{s+2} + s = \frac{s^2 + 2s + 1}{s+2}$$

$$\Rightarrow X(s) = \frac{(s+1)^2}{(s+2)(s-2)(s+1)} = \frac{s+1}{(s+2)(s-2)} = \frac{A}{s+2} + \frac{B}{s-2}$$

$$\text{where } A = \left. \frac{s+1}{(s+2)(s-2)} \cdot (s+2) \right|_{s=-2} = \frac{1}{4}$$

$$B = \left. \frac{s+1}{(s+2)(s-2)} \cdot (s-2) \right|_{s=2} = \frac{3}{4}$$

$$\Rightarrow X(s) = \frac{1}{4} \cdot \frac{1}{s+2} + \frac{3}{4} \cdot \frac{1}{s-2}$$

Now take inverse Laplace transform, we obtain,

$$x(t) = \frac{1}{4} e^{-2t} + \frac{3}{4} e^{2t}$$

Q5 Let A be a square matrix of order n . V be an invertible matrix, prove that A and $V^{-1}AV$ share the same eigenvalues. What can be said about their eigenvectors?

Soln Let λ be an eigen value of A .

$$\lambda \text{ is a root of } \begin{array}{c} \Downarrow \\ |A - \lambda I| = 0 \\ \Uparrow \end{array} \text{ since } V \text{ is invertible } |V| \neq 0$$

$$\lambda \text{ is a root of } |V^{-1}| |A - \lambda I| |V| = 0$$

$$\lambda \text{ is a root of } \begin{array}{c} \Downarrow \\ |V^{-1}A - \lambda V^{-1}I| |V| = 0 \\ \Downarrow \end{array}$$

$$\lambda \text{ is a root of } |V^{-1}AV - \lambda \underbrace{V^{-1}V}_I| = 0$$

$$\lambda \text{ is a root of } \begin{array}{c} \Downarrow \\ |V^{-1}AV - \lambda I| = 0 \\ \Downarrow \end{array}$$

$$\lambda \text{ is an eigenvalue of } V^{-1}AV.$$

on the other hand, let x be an eigenvector of $V^{-1}AV$. Then

$$V^{-1}AVx = \lambda x.$$

$$\Downarrow$$

$$V(V^{-1}AVx) = \lambda Vx$$

$$\Downarrow$$

$$\underline{AVx} = \lambda \underline{Vx}$$

Since x is an eigenvalue of A , Vx is an eigenvector of A . Conversely, if y is an eigenvector of A ,

$$Ay = \lambda y \Rightarrow V^{-1} A y = \lambda V^{-1} y$$

$$\Rightarrow \underbrace{V^{-1} A V}_{V^{-1} A V} \underbrace{V^{-1} y}_z = \lambda \underbrace{V^{-1} y}_z$$

$$\Rightarrow V^{-1} y \text{ is an eigenvector of } V^{-1} A V.$$